

Google's VRP Free Algorithm Evaluation Study and its Use in Small and Medium Transportation Problems

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Abstract—Vehicle routing is one of the oldest, yet critical, problems of the logistics sector. Globalization and the rise of e-commerce as well as technological advances have allowed the development of a profitable routing software market, which has recently attracted the attention of technology giants. Among the plethora of routing software alternatives, Google's open source optimization suite, OR-Tools, looks particularly promising, providing a useful and affordable optimization tool for small and medium sized logistic companies and applications. The goal of this paper is to scientifically evaluate the accuracy, and performance of OR-Tools suite by examining the quality of its solution results in a set of capacitated vehicle routing problems with delivery time windows, also known as capacitated vehicle routing problem with time-windows (CVRPTW) or VRPTW. A comparative analysis is conducted by solving Solomon's VRPTW problems for a small (25 customers), medium (50 customers), and large (100 customers) number of customers with OR-Tools and comparing the results with the known optimal solutions for a fixed maximum solution time and varied number of available vehicles. Additionally, in this paper we demonstrate a real case of application of the OR-Tool tools to a real problem concerning the online distribution of dairy products. But given the very nebulous definition of the online character of a problem, OR-Tool does not include online VRP algorithms. For this reason, we will convert the solution of the classic stationary VRP using OR-Tools into a solution for Online VRP.

Index Terms—VRP, OR Tools, CVRPTW, VRPTW

I. INTRODUCTION

The rapid development of technology and evolution of business models in addition to ever-increasing globalization and the rise of e-commerce have put transportation networks and supply chains under an unprecedented strain. This complicated environment and current environmental concerns make the optimization of the supply chain vital to more and more companies in order to remain competitive, profitable and up-to-date with new, more stringent regulations [1].

Although many technical solutions have been introduced the truth is that most of small and medium sized businesses still rely on instinct, experience, or simplistic methods as the sole basis for their routing practices. A cause of this phenomenon is that the purchase and use of sophisticated routing or

optimization software is often an expensive investment. Using open source optimization software is also challenging because it usually has severe limitations, or it is too complicated to implement in a real-world application. Recently, however, there has been a growing interest in the development of publicly available technology tools, including optimization and routing tools, by big technological giants. An interesting case is the open source optimization suite OR-Tools which is free, simple to use, constantly updated and backed by one of the world's biggest IT companies, Google [2].

The cost differentials between static Vehicle Routing Problems (VRP) and their dynamic counterparts have been the subject of numerous studies. Findings from the literature offer some insight into the average cost increases, despite the fact that the exact percentage increase in objective costs when converting a static VRP to an online VRP can vary. According to research, the conversion of a static VRP to an online VRP results in an increase in operational expenses. For example, the results suggest that the expenses associated with online routing may be somewhat higher than those of static routing, which is indicative of inefficiencies that result from real-time adaptation to changing conditions [3]. In addition the competitive ratio provides an additional viewpoint on cost disparities by contrasting the performance of online algorithms with that of the optimal static solution. This ratio can reveal cost increases ranging from 20% to 40% when transitioning to online systems, depending on situational parameters such as route density and request frequency, according to certain studies [3]. Finally, empirical analyses have revealed scenarios in which online VRPs may incur higher costs as a result of the increased travel distances and delays necessary to accommodate dynamic requests. In environments with significant variability in customer requests, online solutions can result in operational cost increases, as some literature cites a range of 15% to 35% [4]. Nevertheless, it is important to acknowledge that dynamic policies can be more responsive in specific situations, despite the fact that they may result in higher costs in practice as a result of routing fluctuations.

The motivation for this study and its development arose from the collaboration with the local dairy industry, with the aim of optimizing the company's vehicle routing operations. The objective was to study the real-time optimization of a version of the vehicle routing problem, seeking to minimize and determine the transportation cost of the company's products to various retail locations within the region of Magnesia, Greece. In this online model of the vehicle routing problem, new dairy product requests that emerge during the route execution do not affect the distance already traveled but do influence the route's evolution and the sequence of visits to the remaining nodes. The challenge is to determine the best possible route as new demands arise. In practice, this route remains optimal only if no additional requests emerge after its beginning.

II. BACKGROUND AND RELATED WORK

A. The Vehicle Routing Problem

The basic VRP calls for the determination of the optimal set of routes to be performed by a fleet of vehicles to serve a given set of customers from a central depot and is one of the most significant and studied combinatorial optimization problems. The VRP is classified as an NP-hard problem. In transportation and logistics, the VRP presents a major challenges marked by several variants that fit particular operational requirements and constraints. The main groups of VRP variants consist in:

1) *Capacitated Vehicle Routing Problem (CVRP)*: This well-established variant entails the identification of optimal routes for a fleet of vehicles, ensuring that the total demand of served customers does not exceed the capacity of any vehicle [5], [6].

2) *Vehicle Routing Problem with Time Windows (VRPTW)*: In this variation, the routing process is further complicated by the requirement for vehicles to service customers within specific time frames [7].

3) *Vehicle Routing Problem with Pick-up and Delivery (VRPPD)*: This variation is concerned with situations in which vehicles are responsible for the collection and delivery of goods at predetermined locations. In addition to the necessity of adhering to specific routing sequences for both pick-ups and deliveries, constraints encompass vehicle capacities [8].

4) *Heterogeneous Fleet Vehicle Routing Problem (HFVRP)*: The HFVRP effectively models the realistic fleet compositions found in many transport operations, in contrast to the homogeneous fleet models typically considered in traditional VRP frameworks [9].

5) *Multi-Depot Vehicle Routing Problem (MDVRP)*: This variation of the VRP is applicable to situations in which multiple depots contain vehicles that can distribute goods to a variety of customer locations. This complexity introduces additional logistical considerations, as it necessitates the efficient management of routes that originate from multiple depots simultaneously [10].

6) *Open Vehicle Routing Problem (OVRP)*: A distinctive method that eliminates the necessity for vehicles to return to their initial location after delivering goods. This results in

distinct considerations in route optimization and is particularly relevant in situations where the return journey is either economically unfeasible or unnecessary [11].

7) *Routing Issues for Electric Vehicles (E-VRPs)*: E-VRPs are exclusively concerned with the adaptation of conventional VRP frameworks to electric vehicles. They take into account the effects of electric vehicle performance on route efficiency, battery charging times, and driving range [12].

B. Existing solution methods for the VRP

The VRP includes a variety of solution methods that illustrate its complexity and multiple formulations. The methods can be classified into three primary categories: exact algorithms, approximate algorithms, and heuristic/metaheuristic algorithms. Each category provides unique benefits and is appropriate for varying problem scales and specifications. Exact algorithms, by means of exhaustive search of the search space, offer definite solutions for the VRP. These approaches use mathematical programming to identify best paths and consist in linear programming, branch-and-bound, and branch-and-cut techniques. Nevertheless, scalability problems sometimes limit their applicability to small-to medium-sized instances of the problem by their computational intensity. Pisinger and Røpke stress the need of exact techniques for different VRP forms and their shortcomings in effectively handling more significant cases [13]. While it is usually unworkable for complicated VRPs involving many consumers or constraints, this category is helpful when ensuring the best solution is vital [14]. Approximate Algorithms, this group of methods seeks to provide solutions that are sufficiently good but do not guarantee optimality. They often employ a mix of problem-specific heuristics and optimization techniques, designed to quickly yield feasible solutions. A notable method in this category includes the use of greedy algorithms, which construct a solution by iteratively selecting the next best option based on a heuristic criterion [14]. These algorithms are particularly useful when time constraints prioritize rapid solution generation over optimality, especially in dynamic logistical settings.

Heuristic and metaheuristic algorithms have attracted popularity in the literature since they offer strong solutions able to solve more general and more difficult cases of the VRP. While metaheuristics (e.g., genetic algorithms, simulated annealing, tabu search, and ant colony optimization) offer frameworks for more fully exploring the solution space, heuristics operate by producing solutions based on rule-of-thumb strategies without extensive computational overhead. Especially metaheuristics can dynamically change their approaches depending on the performance of past iterations, so improving the quality of the solutions over time [15], [16]. Empirical studies [17] and empirical data [18] have shown, for example, notable success of hybrid algorithms combining several metaheuristic approaches. Their adaptability qualifies them for real-world VRP uses since it combines efficiency with the capacity to manage constraints and variability usually present in logistics operations. As the difficulties in VRP solving have changed, hybrid algorithms—which combine the strengths of exact,

approximative, and heuristic techniques—have surfaced. These approaches combine several algorithms to take use of their respective advantages, such improving computational economy and solution quality. Local search techniques combined with genetic algorithms, for instance, enable adaptive exploration of the solution space and honing of solutions over consecutive iterations [19], [20]. Moreover, hybrid approaches have been successfully applied to maximize new VRP models, including energy-minimizing vehicle routing problems, so demonstrating their adaptability to niche areas inside the VRP domain [15].

III. OR-TOOLS SUITE FOR VRP PROBLEMS

A. OR-Tools

OR-Tools is an open source software suite for optimization, tuned for solving several variants of VRP, flows, integer and linear programming, and constraint programming. OR-Tools API can be used in some of the world's most popular programming languages like C++, C#, Python and Java. OR-Tools suite provides modeling tools for routing and optimization problems and Google's GLOP and CP-SAT mathematical. Third party open-source (SCIP, GLPK) or professional (Gurobi, CPLEX) solvers can also be used parallel to OR-Tools.

With respect to vehicle routing, OR tools are able to solve most of the major VRP variants with no limits on the number of vehicles or customers. The VRP variants covered by OR-Tools are:

- VRP
- CVRP (there is the capability of solving the problem with dropping visits)
- VRPTW (apart from time windows, loading and service times may be considered)
- CVRPTW
- PDP (Pick-up and Delivery Problem, a VRP variation which includes Pick-Ups)

The suite can handle both symmetrical and asymmetrical problems and of homogeneous or not-homogeneous vehicle fleets. It should be noted that the methods used to solve routing problems are not exact, with OR-Tools instead relying on a library of heuristic and metaheuristic algorithms.

Solomon's problems [21] are a set of CVRPTW problems with known optimal solutions, which is often used for benchmarking solution algorithms. Problems are categorized into six subsets according to several factors that affect the behavior of routing and scheduling algorithms.

The name of the subsets consists of two parts. The first part indicates the problem's geographical data structure, with the possible values being R (randomly generated), C (clustered) or RC (randomly clustered), as shown in Fig. 1. The second part indicates low or high maximum capacity vehicles taking the values 1 and 2 accordingly.

Each problem instance consists of 1 depot point and 100 customer points (Euclidean graph), where travel times equal the corresponding distances. For each such problem, smaller problems have been created by considering only the first 25 or 50 customers.

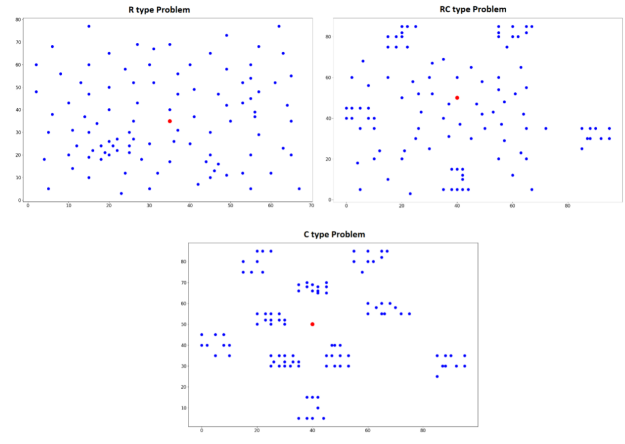


Fig. 1. Different Solomon problem types.

The customer coordinates are identical for all problems within one type (i.e., R, C and RC). The problems differ with respect to the width of the time windows with some of them having very tight time windows, while others have time windows which are hardly constraining.

B. Methodology

In this research the entirety of Solomon's problem set is tested. Python code using OR-Tools Python interface inputs the data for each individual problem, which includes coordinates, demands, time windows, service times and vehicle capacities. Cases are tested with three different numbers of available vehicles equal to the number of vehicles in the known optimal solution plus 50% and 100% additional vehicles (rounded up). The maximum allowed computational time for all problems is fixed to 60 seconds. The table below shows the total number of optimization attempts for each type of problem.

TABLE I
OPTIMIZATION ATEMPS

R1	R2	RC1	RC2	C1	C2	Total
108	99	72	72	81	72	504

Every specific case can have up to three different solutions, one for each number of available vehicles. For the purpose of this research, only the solution with the value closest to the value of the known optimal solution is retained and all others are rejected. If multiple solutions share the same value, the one with the smallest number of available vehicles is selected. A case is deemed unsolved if for every number of available vehicles, a solution cannot be found within 60 seconds.

IV. EXPERIMENTAL RESULTS

A. Small and Medium Transportation Problems

In order to determine the usefulness of OR-Tools suite in real-world application, it was important to examine how well

TABLE II
SOLVED PROBLEMS

Number of customers	R1 type % solved	R2 type % solved	RC1 type % solved	RC2 type % solved	C1 type % solved	C2 type % solved
25	91.67	100	75	100	55.56	100
50	83.33	100	75	100	55.56	62.5
100	91.67	27.27	87.5	62.5	88.89	12.5

TABLE III
AVERAGE PERCENT DEVIATION

Number of customers	R1 type % #	R2 type % #	RC1 type % #	RC2 type % #	C1 type % #	C2 type % #
25	3.11	2.7	4.64	1.87	0.31	1.52
50	5.49	6.95	10.2	4.41	14.28	6.07
100	6.48	6.98	8.58	8.58	10.89	14.22

TABLE IV
AVERAGE EXTRA AVAILABLE AND UTILIZED VEHICLES

Number of customers	R1 % extra vehicles		R2 % extra vehicles		RC1% extra vehicles		RC2% extra vehicles		C1 % extra vehicles		C2 % extra vehicles	
	nv_a	nv_u	nv_a	nv_u	nv_a	nv_u	nv_a	nv_u	nv_a	nv_u	nv_a	nv_u
25	74.5	6.82	39.4	6.82	58.33	5.56	0	0	26.7	0	37.5	0
50	69	9.05	37.22	-3.79	76.67	7.64	39.16	1.13	76	8	60	10
100	92.42	14.95	100	51.39	85.06	15.92	67.78	25.99	87.5	3.75	100	0

TABLE V
DURATION (MIN), TOTAL DISTANCE (M), AND VEHICLE SPEED (KM/H) PER ROUTE IN THE ONLINE VEHICLE ROUTING PROBLEM

route	online time	online dist	online speed	route	online time	online dist	online speed
1	1639	978373	35.81597315	26	1698	720008	25.4419788
2	1612	940924	35.02198511	27	1660	769623	27.8176988
3	932	487478	31.38270386	28	1481	865202	35.05207292
4	1634	724713	26.61124847	29	1719	763171	26.63773124
5	1479	915957	37.15849899	30	1518	579104	22.88948617
6	1261	956944	45.5326249	31	1152	524044	27.29395833
7	1822	864152	28.45725576	32	1614	794186	29.52364312
8	1567	950161	36.38140396	33	1572	707880	27.01832061
9	1619	721374	26.73405806	34	1636	627823	23.0252934
10	1576	875163	33.31838832	35	1556	988002	38.0977635
11	1606	1143185	42.70927771	36	1589	834845	31.52341095
12	1232	502413	24.46816558	37	1043	537488	30.91973154
13	1759	795418	27.1319386	38	1508	816946	32.50448276
14	1508	807543	32.13035809	39	1683	606626	21.62659537
15	1464	802472	32.88819672	40	1039	530570	30.63926853
16	1468	694792	28.39749319	41	1680	665087	23.75310714
17	1704	786367	27.68897887	42	1813	802222	26.54899062
18	1535	533608	20.85764169	43	1533	610505	23.89452055
19	1861	741425	23.90408383	44	1782	816657	27.49686869
20	1683	846440	30.17611408	45	1743	757164	26.06416523
21	1723	893736	31.12255369	46	1379	733111	31.89750544
22	1794	868601	29.05020067	47	1698	875818	30.94763251
23	1418	772363	32.68108604	48	1680	853043	30.46582143
24	1219	664372	32.70083675	49	1494	724937	29.11393574
25	1513	717771	28.46415069	50	1639	643139	23.5438316

could it operate under a strict computational time limit (60 seconds). A good image can be obtained by examining the percentage of cases, where a feasible solution could be found, by problem type and size. The results are presented on Table II and Fig. 2.

As it is expected bigger problems are increasingly harder to solve. For small (25 customers) and medium (50 customers) sized OR-Tools seem to perform well, solving on average more than 75% of the cases, performing better on problems with high capacity vehicles (R2, RC2, C2) and facing difficulties

only in clustered problems with low capacity vehicles (C1 type). The opposite seems to be truth for large (100 customers) problems where the percentage of successfully solved cases drops significantly for high capacity vehicle problems (R2, RC2, C2) while for low capacity vehicle problems it is over 87.5%. The results indicate that structure asymmetrically affects the solvability of different problem types.

Another important factor that must be considered, as OR-Tools rely mostly on heuristics and metaheuristics, is the precision of the optimization process, which indicates how

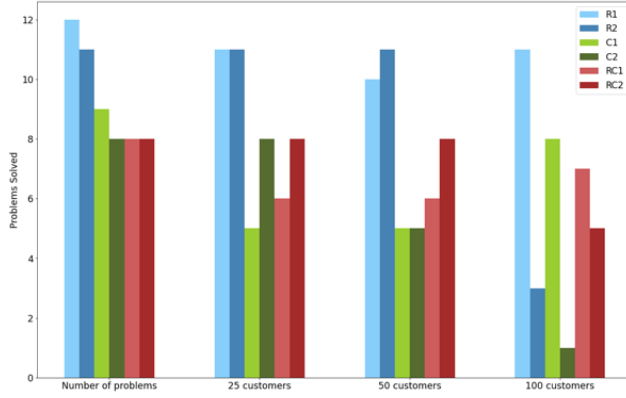


Fig. 2. Problems solved by type and size.

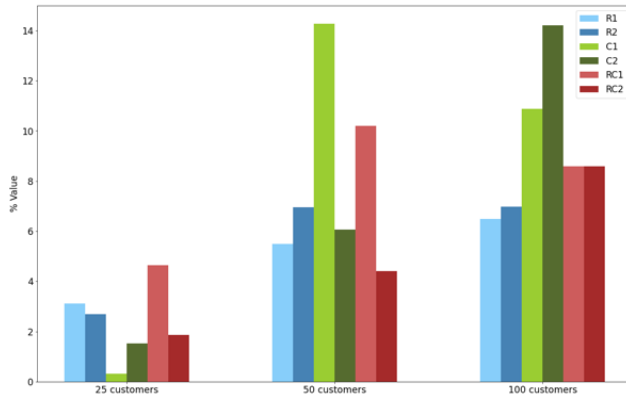


Fig. 3. Average % deviation by type and size.

close is the calculated solution to the actual mathematical optimal solution. A deviation percent error (e_d) between the calculated (v_c) and the known optimal (v_k) solution values can be calculated using the formula $e_d = 100((v_c - v_k)/v_k)$. The average e_d value by problem type and size is presented on Table III and Fig. 3. The average e_d is calculated excluding the unsolved cases.

The results show that e_d is below 15% for all the cases and for the majority of cases below 10%. As it is expected, larger problems are harder to solve, thus the increase of problem size correlates to the increase in the average e_d . In large problems (100 customers) OR-Tools seem to perform better in problems with random structure (R type). As the problem structure becomes more clustered an increase in the average e_d is observed.

As previously mentioned, every case was tested for 3 different numbers of available vehicles, 0%, 50% and 100% additional vehicles from the known optimal solution. Even though sometimes OR-Tools require more available vehicles to successfully find a solution they don't necessarily utilize all of them. The average percentage of extra available vehicles needed (nv_a) for the OR-Tools to find a solution and the

average percentage of extra vehicles utilized (nv_u) in OR-Tools solution compared to the vehicles in the known optimal solution by problem type and size is presented on Table IV. The average values are calculated excluding the unsolved cases.

The results indicate that even the nv_a values are generally high the values of nv_u stay relatively low. This is positive because it means that if CVRP is hard to solve given a number of available vehicles, the strategy of increasing the number of available vehicles to make the problem easier to solve is valid because although the number of extra vehicles may be high the number of actual used vehicles will be close to the number of the starting available vehicles.

B. Problem definition for our real life test Case

A single transport vehicle is utilized, with sufficient capacity to fulfill all demands. Initially, the model computes an optimal route visiting all retail locations (a total of 30), starting from the company's facilities (DEPOT – node 0), based on the input data (temporal sequence of demands and locations of retail points). As the vehicle visits a number of locations, a new demand for products arises, prompting the model to recalculate an optimized route that incorporates the new demand alongside the remaining unfulfilled ones. This process repeats as new demands emerge until all retail locations are visited. The demand study timeframe is divided into four time windows (0-120, 120-240, 240-360, 360-480 minutes) – (08:00-10:00, 10:00-12:00, 12:00-14:00, 14:00-16:00). Each new request is considered for fulfillment from the subsequent time window following its emergence. The waiting time for each node's demand fulfillment was set to be sufficiently large to ensure that, in the event of multiple simultaneous requests within the same time window due to randomness, the results remain unaffected. The selection of both the number of submodels within the method (with a maximum of 3) and the number of retail points included in each submodel, as well as the time windows in which respective demands are met is random, following the steps below:

- A number between 1 and 30 is selected, indicating the number of nodes to be served in the first submodel's resolution.
- A set of numbers between 1 and 30 is chosen, corresponding to the number selected in step (a). These nodes will be served in the first submodel's resolution.
- For each node selected in step (b), a number between 1 and 4 is chosen, indicating the time window in which it will be served.
- A number between 1 and [30 - (number from step (a))] is selected, determining the number of nodes to be served in the second submodel's resolution.
- A set of numbers between 1 and 30, corresponding to the number selected in step (d), is chosen. These nodes will be served in the second submodel's resolution, excluding those already considered in the first submodel.
- For each node selected in step (e), a number between 2 and 4 is chosen, indicating the time window in which

it will be served. The number 1 (first time window) is excluded, as the route has already started, and new requests can only be fulfilled from the second time window onwards.

- (g) A number between 1 and $[(30 - (\text{number from step (a)}) - (\text{number from step (d)}))]$ is selected, determining the number of nodes to be served in the third submodel resolution.
- (h) A set of numbers between 1 and 30, corresponding to the number selected in step (g), is chosen, again excluding nodes already considered in the previous submodels.
- (i) For each node selected in step (h), a number between 3 and 4 is chosen, indicating the time window in which it will be visited by the vehicle. Any remaining numbers between 1 and 30 that have not been selected in steps (b), (e), and (h) are the nodes not assigned to any time window and will be examined in the fourth and final one.

In the first submodel, the initial node (DEPOT) is set as 0. In the second, the initial node is selected as the last visited node from the first submodel's route within the first time window. By solving these subproblems, the total travel time of the routes is calculated, and by summing the individual arc distances, the total distance traveled by the vehicle is also determined, both for each submodel and the entire online model. A total of 50 models were studied. Based on the adopted methodology, the total transportation times of 50 dynamic models of the vehicle routing problem were determined, along with the corresponding times of each model's submodels. The automatic solution strategy was applied in combination with the automatic improvement algorithm. Additionally, given the distances between nodes, the traveled distance of each submodel and the total model distance were calculated.

C. Numerical Results

By analyzing the solution results, a comparison was made between the behavior of the online and the offline system. In the offline system, all demands are known from the beginning of the problem, with no new requests emerging during the route execution. All demands are placed in chronological order and can only be fulfilled from the subsequent time window onward. The travel times of the static and dynamic versions were compared (Fig. 5), along with their respective total distances (Fig. 4). The numerical data, as well as the average vehicle speed for each model, are presented in Tables V and VI. The total travel time of the online version was defined as the sum of the individual submodels' times.

Based on the results, there is a significant difference in the travel times, and as expected, the solution degrades in the case of the online model. Knowledge of all the problem data from the outset seems to facilitate the modeling process, yielding better results. An exception is the third model, for which the two examined times are identical (932 minutes), and consequently, their standard deviation is zero. This can be explained by the fact that during the submodel definition process, all nodes were designated to be examined in the first submodel, making it the only submodel for the third

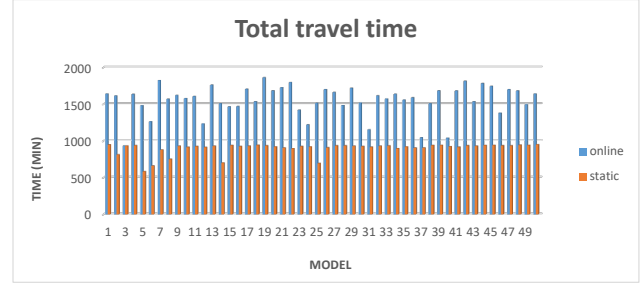


Fig. 4. Diagram comparing travel times of the static and dynamic model.

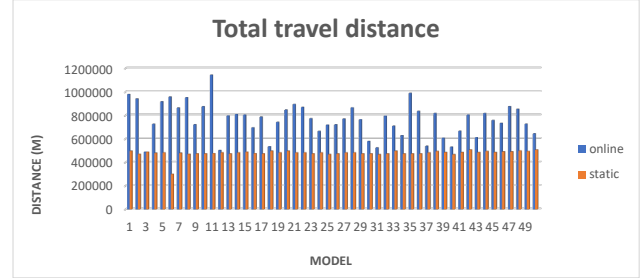


Fig. 5. Diagram comparing travel distance of the static and dynamic model.

model, with no difference from the corresponding static model. Furthermore, the standard deviation appears to be significantly smaller for the entirety of the observations in models 37 and 39, where the dynamic version has two submodels instead of three. This occurred because, during the selection process for the submodel elements, in the second submodel of each case, the examined nodes were all those not considered in the first.

V. CONCLUSIONS

The outcome of comparative analysis shows that OR-Tools can be useful and accurate enough to meet the needs of small and medium sized CVRPTW applications. Small (25 customers) and medium (50 customers) sized problems were solved on average over 75% of the time. For large problems (100 customers), cases with small capacity vehicles were solved on average over 87.5% of the time, while cases with high capacity vehicles were difficult to solve. The quality of the solutions produced, for most cases, was found to be good with average deviation less than 10% from the known optimal solutions. Finally, it was observed that a significant increase in the number of vehicles available, as inputs, was required to obtain solutions, but most of the additional vehicles were not used and the actual number of vehicles needed was less than 30% of the number of vehicles in the optimal solution. The results were obtained within strict time limits, which suggests that OR-Tools could be used reliably in real-world problems. Moreover, based on the above real world use case, the static model, in which all necessary data are known before the start of the route, seems to demonstrate better behavior and performance.

TABLE VI
DURATION (MIN), TOTAL DISTANCE (M), AND VEHICLE SPEED (KM/H) PER ROUTE IN THE OFFLINE VEHICLE ROUTING PROBLEM

route	static time	static dist	static speed	route	static time	static dist	static speed
1	948	495426	31.35607595	26	908	472610	31.22973568
2	812	469910	34.72241379	27	934	481699	30.94426124
3	932	487478	31.38270386	28	934	481699	30.94426124
4	937	477807	30.59596585	29	929	472610	30.52378902
5	583	481725	49.57718696	30	925	472610	30.65578378
6	662	298960	27.09607251	31	916	466140	30.53318777
7	876	477807	32.72650685	32	929	472533	30.51881593
8	751	469910	37.54274301	33	933	495425	31.86012862
9	929	472533	30.51881593	34	896	472533	31.64283482
10	915	472533	30.98577049	35	919	472610	30.85593036
11	925	475324	30.83182703	36	903	472533	31.39754153
12	912	482144	31.72844668	37	905	481699	31.9358453
13	929	472533	30.51881593	38	939	493524	31.53507987
14	700	481725	41.29071429	39	941	484058	30.86448459
15	939	487478	31.14875399	40	923	466140	30.30162514
16	925	475324	30.83182703	41	916	484058	31.7068559
17	929	472610	30.52378902	42	935	505603	32.4451123
18	942	495829	31.58146497	43	929	483168	31.20568353
19	934	481699	30.94426124	44	940	493600	31.50638298
20	919	495425	32.34548422	45	941	484058	30.86448459
21	905	481699	31.9358453	46	935	490445	31.47240642
22	894	481775	32.33389262	47	934	489453	31.44237687
23	925	472610	30.65578378	48	943	498464	31.71563097
24	919	481699	31.44933624	49	941	493313	31.45460149
25	693	467119	40.44320346	50	950	505603	31.93282105

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